# HBG RAG-DOMINANCE: IaC PACKAGE

"Никогда не доверяйте человеку то, что может исполнить алгоритм."

## /iac/access_matrix.yaml


```yaml
# HBG RAG-DOMINANCE: CONTROL MATRIX
# Version: 1.1 (Machiavelli Edition)
# Purpose: Total Credit Risk Elimination via 6x11 Framework

project_id: "hbg-rag-dominance"
region: "europe-central2"

# 11 Stages of Truth (Process Mapping)
stages:
  1: "Ingestion (U4)"
  2: "Normalization (U4)"
  3: "Embedding (U3)"
  4: "Indexing (U4)"
  5: "Orchestration (U2)"
  6: "Retrieval (U3)"
  7: "Reasoning (U3)"
  8: "Risk Scoring (U1/U3)"
  9: "Verification (U5 - Inquisitor)"
  10: "Persistence (U2/U1)"
  11: "Audit Trace (U6)"

# The 6 Agents of the System
agents:
  - id: "u1_architect"
    email: "ok-admin@gmail.com"
    clearance: "GrandMaster"
    gcp_roles: ["roles/owner"]
    k8s_access: "cluster-admin"

  - id: "u2_sre"
```

```yaml
    email: "gw-devops@gmail.com"
    clearance: "Execution_Lead"
    gcp_roles: ["roles/editor", "roles/container.admin"]
    k8s_access: "edit"

  - id: "u3_ml_designer"
    email: "ux-dev@gmail.com"
    clearance: "Cognitive_Designer"
    gcp_roles: ["roles/aiplatform.user", "roles/storage.objectViewer
    k8s_access: "edit"

  - id: "u4_data_custodian"
    email: "sh-dev@gmail.com"
    clearance: "Resource_Master"
    gcp_roles: ["roles/storage.objectAdmin", "roles/bigquery.dataEdi
    k8s_access: "view"

  - id: "u5_inquisitor"
    email: "pk-qa@gmail.com"
    clearance: "Truth_Verifier"
    gcp_roles: ["roles/viewer"]
    k8s_access: "edit"

  - id: "u6_auditor"
    email: "ok-audit@gmail.com"
    clearance: "Legal_Shield"
    gcp_roles: ["roles/bigquery.metadataViewer", "roles/viewer"]
    k8s_access: "view"
```

## /iac/main.py (Pulumi Python)

```python
import pulumi
import pulumi_gcp as gcp
import pulumi_kubernetes as k8s
import yaml

# Load the Digital Code of Conduct
with open("access_matrix.yaml", "r") as f:
    matrix = yaml.safe_load(f)

project = matrix['project_id']
region = matrix['region']
```

```
# 1. Create Storage Buckets for the 11-Stage Workflow
for env in ["dev", "ref", "prod"]:
    bucket = gcp.storage.Bucket(f"hbg-data-{env}",
        location=region,
        uniform_bucket_level_access=True)

# 2. Distribute Sovereign Rights (GCP IAM)
for agent in matrix['agents']:
    for role in agent['gcp_roles']:
        gcp.projects.IAMMember(f"iam-{agent['id']}-{role.replace('.'
            project=project,
            role=role,
            member=f"user:{agent['email']}")

# 3. K8s RBAC: Enforcing the 6-Agent Hierarchy
for agent in matrix['agents']:
    for env in ["dev", "stage", "prod"]:
        ns = f"camunda-{env}"
        k8s.rbac.v1.RoleBinding(f"rb-{agent['id']}-{env}",
            metadata={"namespace": ns},
            role_ref=k8s.rbac.v1.RoleBindingRoleRefArgs(
                api_group="rbac.authorization.k8s.io",
                kind="ClusterRole",
                name=agent['k8s_access']
            ),
            subjects=[k8s.rbac.v1.RoleBindingSubjectArgs(
                kind="User",
                name=agent['email']
            )])

pulumi.export("system_status", "DOMINANCE_ESTABLISHED")
```